

Effects of date (*Phoenix dactylifera* L., Medjool or Hallawi Variety) consumption by healthy subjects on serum glucose and lipid levels and on seru...

.

The present pilot study analyzed, for the first time, the in vivo effect of Medjool or Hallawi date consumption by healthy subjects on serum glucose, lipids, and oxidative stress. Total phenolics concentration in the Hallawi versus Medjool dates was greater by 20-31%. The major proportion of the soluble phenolics in both date varieties consisted of phenolic acids, mainly ferulic acid and coumaric acid derivatives, and also chlorogenic and caffeic acid derivatives. Unlike the Medjool dates, Hallawi dates contained a significant proportion of catechins as well. In addition, both varieties contained a quercetin derivative. Both date varieties possess antioxidative properties in vitro, but the ferric ion reducing antioxidant power of Hallawi versus Medjool dates was higher by 24%. Ten healthy subjects consumed, for a period of 4 weeks 100 g/day of either Medjool or Hallawi dates. The date consumption did not significantly affect the subjects' body mass index (BMI), their serum total cholesterol, or their cholesterol levels in the VLDL, LDL, or HDL fractions. Most important, fasting serum glucose and triacylglycerol levels were not increased after consumption of either date variety, and serum triacylglycerol levels even significantly ($p < 0.05$) decreased, by 8 or 15% after Medjool or Hallawi date consumption, respectively. Basal serum oxidative status was significantly ($p < 0.01$) decreased by 33%, as compared to the levels observed before consumption, after Hallawi (but not Medjool) date consumption. Similarly, the susceptibility of serum to AAPH-induced lipid peroxidation decreased by 12%, but only after Hallawi date consumption. In agreement with the above results, serum activity of the HDL-associated antioxidant enzyme paraoxonase 1 (PON1) significantly increased, by 8%, after Hallawi date consumption. It is concluded that date consumption (and mainly the Hallawi variety) by healthy subjects, despite their high sugar content, demonstrates beneficial effects on serum triacylglycerol and oxidative stress and does not worsen serum glucose and lipid/lipoprotein patterns, and thus can be considered an antiatherogenic nutrient .